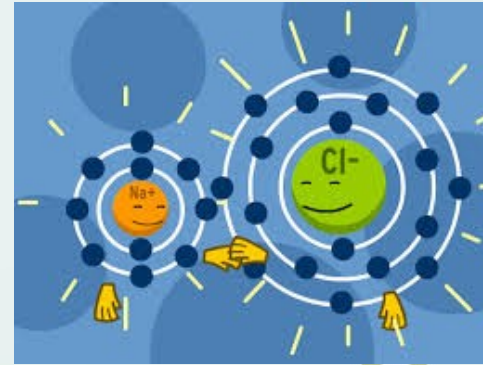
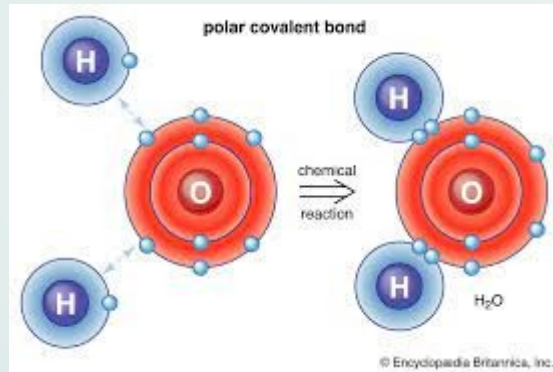


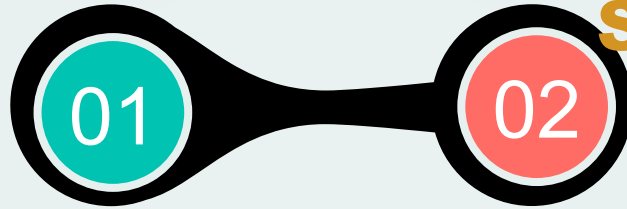
# The Effect of Bond Types on the Structures of Solid Compounds - Molecular Models(pg 20-22)



# Slides for Presentation

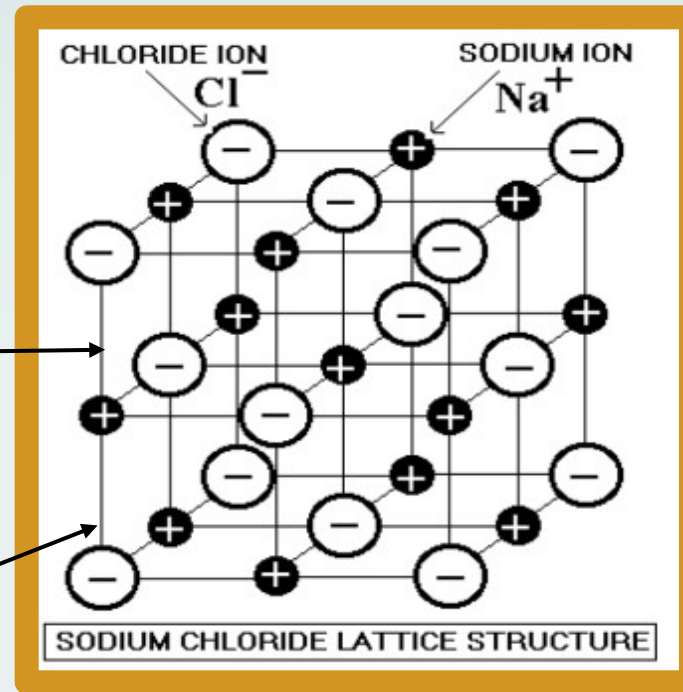
**Covalent  
Network or  
Molecular  
Solid**

**Ionic  
Lattice**



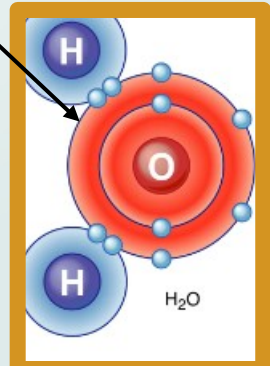
# Ionic Lattice

- electrostatic forces created by the positive and negative charges (opposites attract) in an ionic bond extend in all directions
  - compounds with ionic bonds exist as lattice networks extending in 3D
  - Example: sodium chloride, which consists of a lattice of alternating + sodium and - chlorine ions [chlorine stole some electrons from sodium] repeated until the edge of a salt crystal is reached



# Covalent Network or Molecular Solid

- The force creating a covalent bond is focused on just the two atoms involved
  - The atoms share electrons and that makes them stick together (people sharing internet in houses)
    - while hydrogen only forms a single covalent bond in the compound, carbon can form four covalent bonds with other atoms
  - Covalently bonded molecules can exist as small units, such as methane and water, or as a covalent network, like diamonds, which consists of carbon, each atom bonds to four other carbon atoms in a lattice of such bonds, forming a diamond (streets connecting other roads, houses, or highway)





# EXTRA SLIDE: METALLS



# Bonds

- ionic
- covalent
- metallic

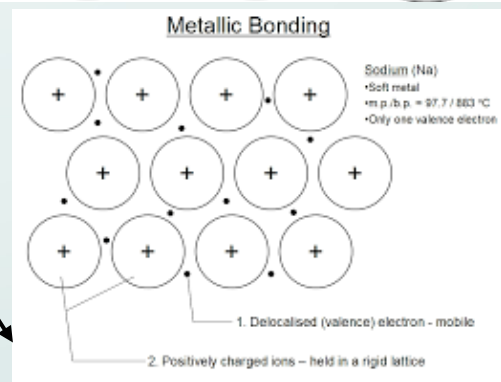
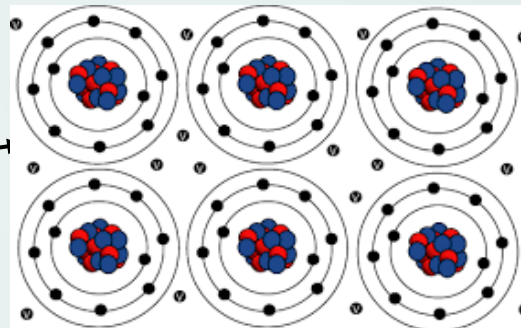
Metals - conduct electricity.  
malleable "solid" at room temp.

Non metals

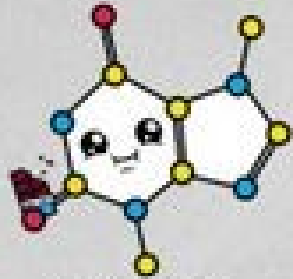
metal x 2  
metal - non-metal - ionic bonds  
non-metal x 2 - covalent

|          |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           |          |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |          |          |         |         |          |          |
|----------|----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------|----------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|----------|----------|---------|---------|----------|----------|
| 1<br>H   |          |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 2<br>He   |          |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |          |          |         |         |          |          |
| 3<br>Li  | 4<br>Be  |          |           |           |           |           |           |           |           |           |           |           |           |           |           |           |           | 11<br>Na | 12<br>Mg |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  | 13<br>Al | 14<br>Si | 15<br>P | 16<br>S | 17<br>Cl | 18<br>Ar |
| 19<br>K  | 20<br>Ca | 21<br>Sc | 22<br>Ti  | 23<br>V   | 24<br>Cr  | 25<br>Mn  | 26<br>Fe  | 27<br>Co  | 28<br>Ni  | 29<br>Cu  | 30<br>Zn  | 31<br>Ga  | 32<br>Ge  | 33<br>As  | 34<br>Se  | 35<br>Br  | 36<br>Kr  |          |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |          |          |         |         |          |          |
| 37<br>Rb | 38<br>Sr | 39<br>Y  | 40<br>Zr  | 41<br>Nb  | 42<br>Mo  | 43<br>Tc  | 44<br>Ru  | 45<br>Rh  | 46<br>Pd  | 47<br>Ag  | 48<br>Cd  | 49<br>In  | 50<br>Sn  | 51<br>Sb  | 52<br>Te  | 53<br>I   | 54<br>Xe  |          |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |          |          |         |         |          |          |
| 55<br>Cs | 56<br>Ba | 57<br>La | 72<br>Hf  | 73<br>Ta  | 74<br>W   | 75<br>Re  | 76<br>Os  | 77<br>Ir  | 78<br>Pt  | 79<br>Au  | 80<br>Hg  | 81<br>Tl  | 82<br>Pb  | 83<br>Bi  | 84<br>Po  | 85<br>At  | 86<br>Rn  |          |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |          |          |         |         |          |          |
| 87<br>Fr | 88<br>Ra | 89<br>Ac | 104<br>Rf | 105<br>Db | 106<br>Sg | 107<br>Bh | 108<br>Hs | 109<br>Mt | 110<br>Ds | 111<br>Rg | 112<br>Cn | 113<br>Nh | 114<br>Fl | 115<br>Mc | 116<br>Lv | 117<br>Ts | 118<br>Og |          |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |          |          |         |         |          |          |
| 58<br>Ce | 59<br>Pr | 60<br>Nd | 61<br>Pm  | 62<br>Sm  | 63<br>Eu  | 64<br>Gd  | 65<br>Tb  | 66<br>Dy  | 67<br>Ho  | 68<br>Er  | 69<br>Tm  | 70<br>Yb  | 71<br>Lu  |           |           |           |           |          |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |          |          |         |         |          |          |
| 90<br>Th | 91<br>Pa | 92<br>U  | 93<br>Np  | 94<br>Pu  | 95<br>Am  | 96<br>Cm  | 97<br>Bk  | 98<br>Cf  | 99<br>Es  | 100<br>Fm | 101<br>Md | 102<br>No | 103<br>Lr |           |           |           |           |          |          |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |          |          |         |         |          |          |

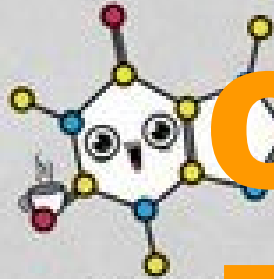
- Metals easily lose electrons
- Metallic bonds are highly ordered solids with metal atoms with free-flowing electrons
- They form a grid of atoms that, have electrons hold the atoms tightly
  - Similar to how ionic bonds form lattice structures
    - Makes metals strong, but also can change shape as layers of the atoms can slide over one another relatively easily.



Chemistry is awesome



Chocolate  
(Theobromine)



Caffeine



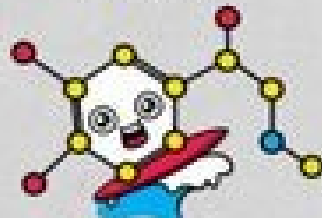
Alcohol  
(Ethanol)



Happiness  
(Serotonin)



Love  
(Dopamine)



Adrenaline

Thank You for  
Coming to My  
Presentation!